

Band Theory of Solids

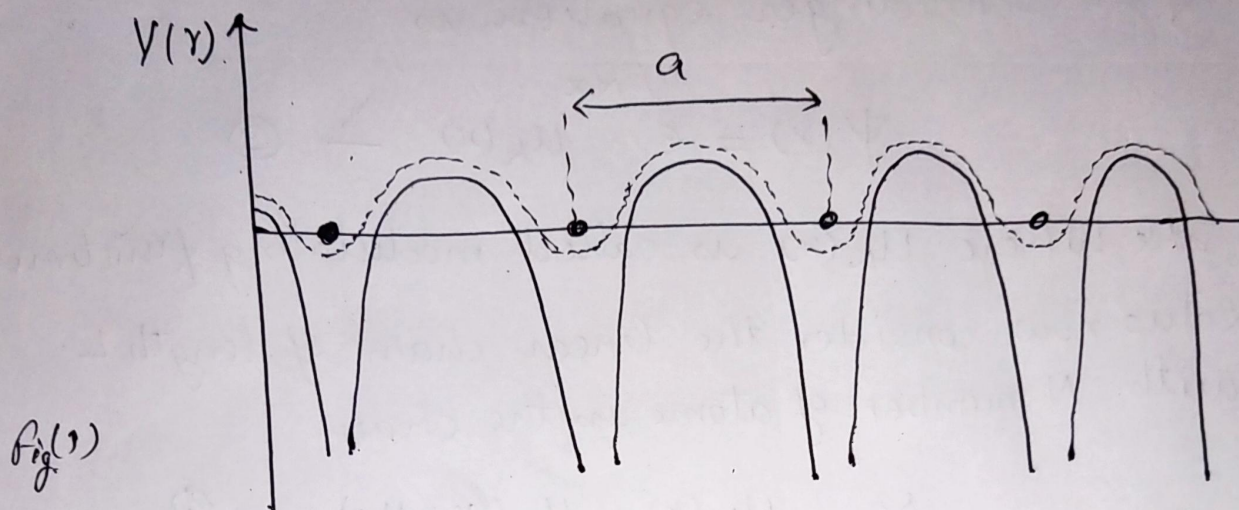


Fig (1)

Bloch Theorem : Metals are crystalline in nature.

The potential at the positive ion ~~core~~ site is minimum and maximum between the two ions. The conduction electrons move inside the positive ion cores and therefore electron moves in a periodic potential as shown in the above figure.

Suppose the periodicity of the crystal lattice is a . Therefore the periodic potential can be defined as

$$V(x) = V(x+a) \quad \text{--- (1)}$$

Suppose the electron energy while moving in a periodic potential is E .

Thus the Schrodinger equation for an electron in a periodic potential can be written as

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} (E - V) \psi = 0 \quad \text{--- (2)}$$

④

Block has given the one dimensional solution of the Schrodinger equation is

$$\psi(x) = e^{i k x} u_k(x) \quad \text{--- (3)}$$

Where $u_k(x)$ is called modulating function. Let us now consider the linear chain of length L with N number of atoms in the chain.

$$\text{so } u_k(x) = u_k(x + Na) \quad \text{--- (4)}$$

From equations (3) & (4), we can write

$$\psi(x + Na) = e^{i k (x + Na)} u_k(x + Na)$$

$$\Rightarrow \psi(x + Na) = e^{i k x} e^{i k Na} u_k(x)$$

$$\Rightarrow \boxed{\psi(x + Na) = \psi(x) e^{i k Na}} \quad \text{--- (5)}$$

Complex conjugate of equation (5)

$$\psi^*(x + Na) = \psi^*(x) e^{-i k Na} \quad \text{--- (6)}$$

(3)

Now

$$\psi^*(x+Na) \psi(x+Na) = \psi^*(x) \psi(x)$$

$$\boxed{|\psi(x+Na)|^2 = |\psi(x)|^2} \quad - (7)$$

⇒ Equation (7) refers that electron is not localized around one particular atom but the probability of finding the electron is same throughout the crystal.

Therefore from equation (5), we

$$\frac{\psi(x+Na)}{\psi(x)} = e^{iNka} = 1$$

$$\Rightarrow e^{iNka} = e^{i2n\pi}$$

$$\Rightarrow Nka = 2n\pi$$

where ~~n = 0, \pm 1, \pm 2, \dots~~
 $n = \pm 1, \pm 2, \pm 3, \dots$

$$\Rightarrow k = \frac{2n\pi}{Na}$$

$$\Rightarrow k = \frac{2n\pi}{L}$$

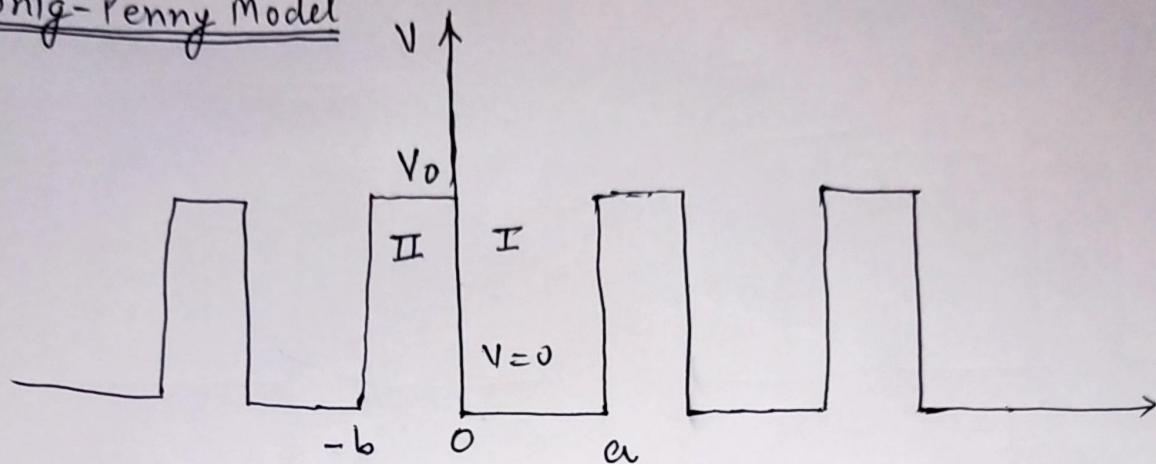
as we assumed earlier.

$$L = Na$$

⇒ L is length of the lattice chain.

Kronig-Penny Model

(4)



Kronig & Penny introduced a simpler model for the shape of potential variation shown in above figure. The potential variation shown in above figure is rectangular in shape and the potential values in different regions is as follows.

$$V = 0 \quad \text{for } 0 < x < a$$

$$V = V_0 \quad \text{for } -b < x < 0$$

Suppose electron's ~~total~~ energy is given by E . Therefore the wave function associated with electron can be calculated by solving Schrodinger equation. Schrodinger equation in region I

$$\frac{d^2 \psi}{dx^2} + \frac{2mE}{\hbar^2} \psi = 0 \quad \text{for } 0 < x < a$$

— (1)

(5)

Schrodinger equation in region II

$$\frac{d^2\psi}{dx^2} + \frac{2m(E-V)}{\hbar^2} \psi = 0 \quad - (2) \text{ for } -b < x < 0$$

Suppose

$$\alpha = \sqrt{\frac{2mE}{\hbar^2}}$$

$$\beta = \sqrt{\frac{2m(V_0-E)}{\hbar^2}}$$

Equations ① & ② can be written as.

$$\frac{d^2\psi}{dx^2} + \alpha^2 \psi = 0 \quad - (3) \text{ for } 0 < x < a$$

$$\frac{d^2\psi}{dx^2} - \beta^2 \psi = 0 \quad - (4) \text{ for } -b < x < 0$$

Solution of the equation ③ & ④ can be written as using Bloch theorem

$$P \frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka \quad - (5)$$

where $P = \frac{mV_0ab}{\hbar^2} \quad - (6)$

⇒ Where $V_0 b$ is called the barrier strength. ⑤

and $\rho = \frac{m V_0 a b}{\hbar^2}$ is called the scattering power of the potential barrier. It is a measure of the strength with which electrons in the ~~electrons~~ crystal are scattered to the ions on the crystal lattice sites.

Further since

$$\alpha^2 = \frac{2mE}{\hbar^2}$$

$$\Rightarrow E = \frac{\hbar^2 \alpha^2}{2m}$$

$$K = \frac{2\pi}{\lambda}$$

In equation ⑤ there are only two variables α & K .

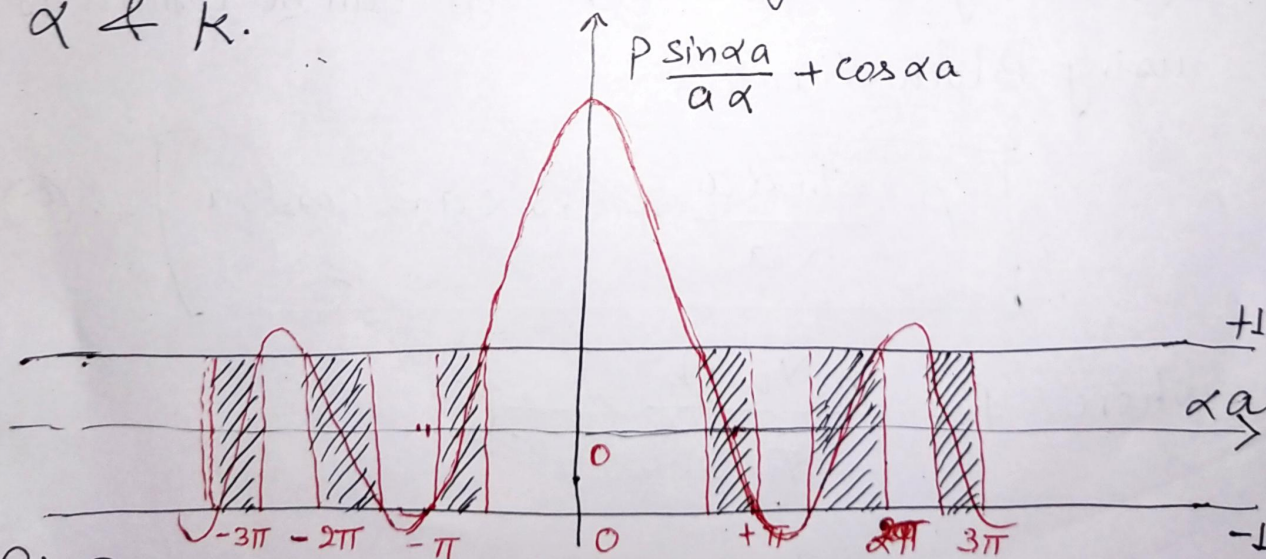


Fig ②

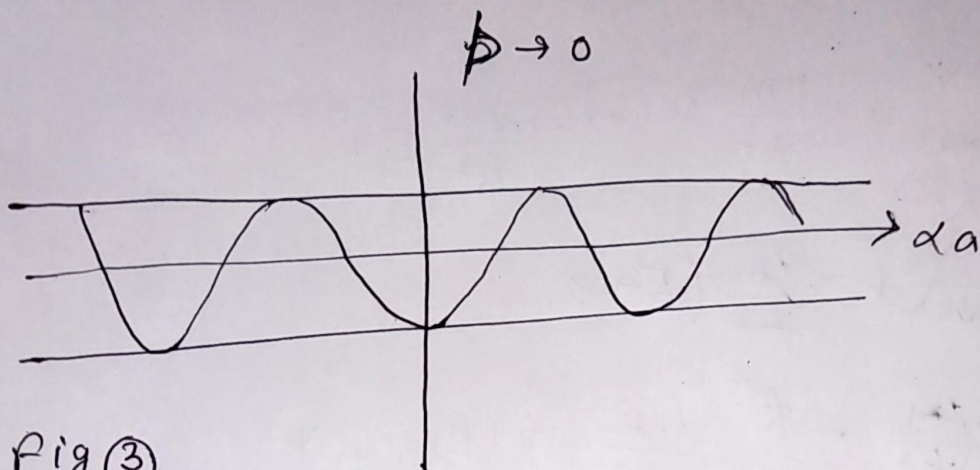


Fig (3)

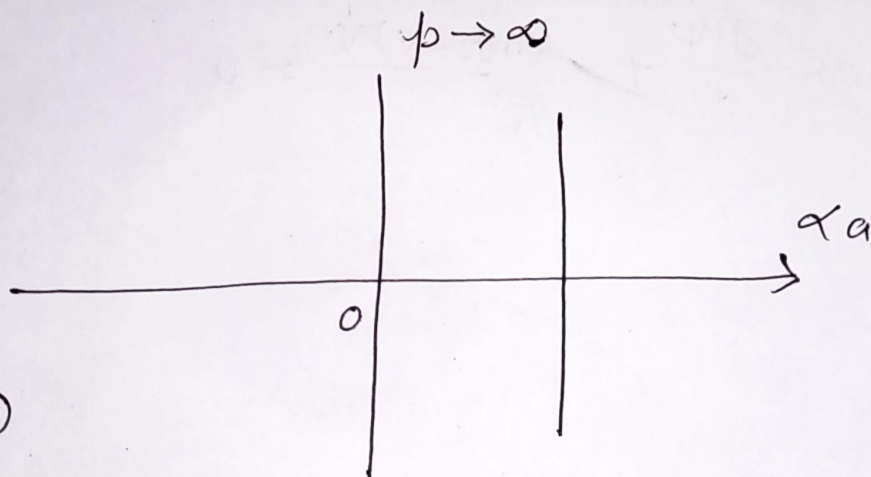


Fig (4)

From fig. (3) only some of the range of αa values are allowed indicating the limiting range of energies are allowed. Dark region indicates the allowed energies & white region is forbidden.

Special cases.

(i) If $P = \frac{m V_0 a b}{\hbar^2} \rightarrow 0$

From equation (5)

$$\cos \alpha a = \cos ka.$$

$$\Rightarrow K = \alpha$$

$$\Rightarrow K^2 = \alpha^2 = \frac{2mE}{\hbar^2}$$

$$\Rightarrow \boxed{E = \frac{\hbar^2 K^2}{2m} = \frac{p^2}{2m}} \quad \text{--- (7)}$$

$$\text{Since } E = K.E. + P.E. \quad \text{--- (8)}$$

Comparing equation (7) & (8)

$$P.E. = 0$$

$$\Rightarrow V = 0$$

$\Rightarrow$ This is the case of free electrons.

(ii) If $p \rightarrow \infty$, the allowed bands reduces to a single energy level. This is the special case when electron trapped in a ~~single~~ potential well.

Conclusion : $\rightarrow$ (i) Electrons in solids are permitted to be in allowed energy bands separated by forbidden energy gaps.

(ii) Allowed energy band width increases with αa .

(iii) $p \rightarrow \infty$ is the case of electron trapped & $p \rightarrow 0$ is the case of free particle.

①

BRILLOUIN ZONES

Since we know that from KRONIG PENNEY model, schrodinger wave equation solutions

$$\frac{d^2 \psi}{dx^2} + \alpha^2 \psi = 0 \quad \text{--- (1)} \quad 0 < x < a$$

$$\frac{d^2 \psi}{dx^2} - \beta^2 \psi = 0 \quad \text{--- (2)} \quad -b < x < 0$$

where $\alpha = \sqrt{\frac{2mE}{\hbar^2}}$

$$\beta = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}}$$

$$P \frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka \quad \text{--- (3)}$$

where $P = \frac{m V_0 a b}{\hbar^2}$

since from equation (3)

$$\cos ka = \pm 1$$

~~$$ka = \frac{n\pi}{2}$$~~

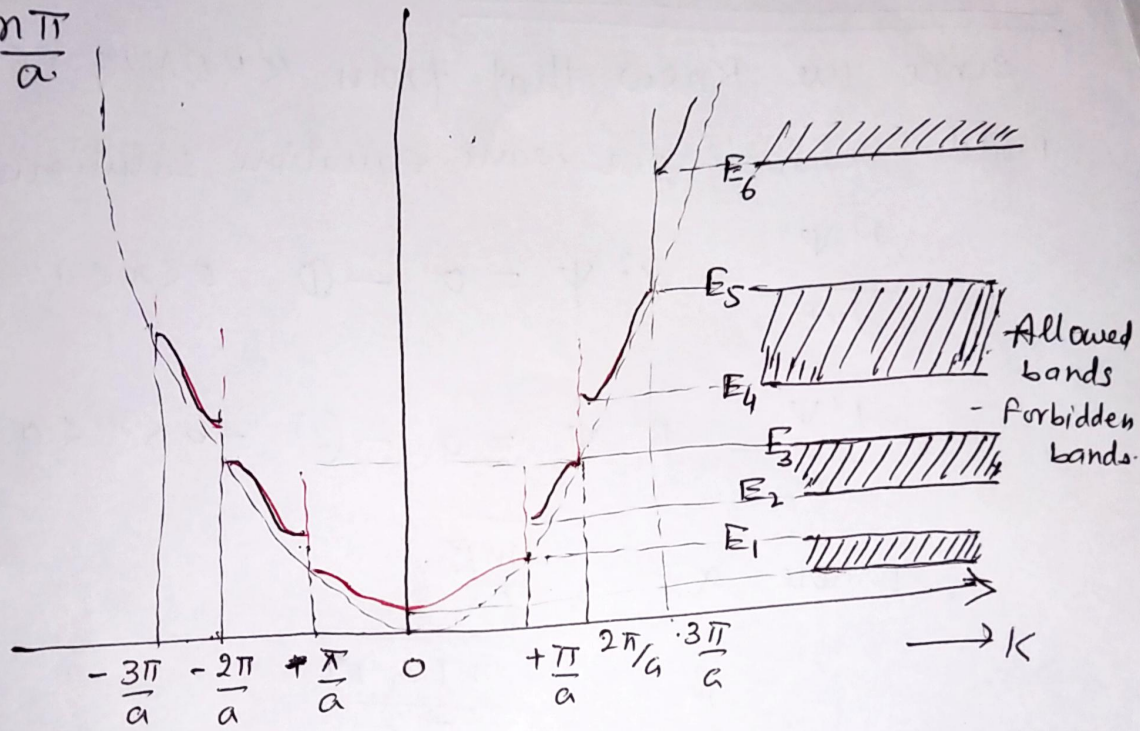
$$\Rightarrow ka = n\pi$$

$$k = \frac{n\pi}{a} \quad \text{--- (4)}$$

where $n = \pm 1, \pm 2, \pm 3,$

The discontinuities in E & K graph occurs at

$$K = \frac{n\pi}{a}$$



(2)

E k diagram

Since the motion of free electron inside the crystal can be written as in terms of kinetic energy

$$E = \frac{\hbar^2 k^2}{2m} \quad \text{--- (1)}$$

where $p = \hbar k$ is the true momentum of the electron.

However electron moving in a periodic potential, $p = \hbar k$ does not represent its true momentum

~~velocity~~ Velocity of the electron inside the crystal

$$v = \frac{1}{\hbar} \left(\frac{dE}{dk} \right) \quad \text{--- (2)}$$

Effective mass of the electron is given by

$$m^* = \frac{\hbar^2}{\left(\frac{d^2 E}{dk^2} \right)}$$

